

Summary: Incomplete Work

Inside Airbnb - New York City | Supplementary Deliverable (Assignment Section 11.1)

The assignment brief states explicitly that it is scoped beyond what one person can complete in one week, and that the evaluation rewards honest prioritisation over superficial completeness. This document is the candid account the brief asks for: what was deliberately not built, why it was deprioritised, and what would be done with more time. Nothing below should be read as a missed deadline -it reflects a prioritisation decision made on day one and revisited throughout the week (see companion *Assumptions & Decisions Log*, Section A.2).

Prioritisation Philosophy

The priority ordering applied was: (1) Section 02, Dataset Familiarization - mandatory and foundational; (2) Section 03, Data Engineering Challenges -the single highest-weighted rubric item (25%); (3) Section 04, Exploratory Data Analysis - the next-highest combined weight, and the section that converts engineering work into business value. Everything in Sections 03.6 and 05-08 was consciously deferred so that Sections 02–04 could be completed to a high standard rather than every section being attempted shallowly.

What Was Not Completed

Section	Status	What's Missing	Why Deprioritised
3.6 - Advanced & Cloud-Native Topics	Not attempted	Docker/docker-compose deployment, 50+ city scaling design, calendar partitioning strategy, change-data-capture (CDC) implementation, cloud-native (AWS/GCP/Azure) deployment, data-quality-as-code (e.g. Great Expectations/dbt tests).	Expert difficulty, Optional category, 6–12h estimated effort. The core engineering pipeline (3.1–3.5) and EDA were mandatory/recommended and rubric-weighted higher; architectural trade-offs for this topic are discussed narratively in the report instead (Section 5, Engineering Approach).
5 -Statistical Analysis	Partially documented	Full H1–H5 hypothesis-testing workflow (explicit null/alternative statements, assumption checks, confidence intervals, and effect sizes computed end-to-end across all 222 neighbourhoods and the full calendar weekday/weekend comparison) was not run to completion. The report presents the methodology and expected/indicative results rather than fully verified statistics for H1, H3, H4 and H5.	Intermediate–Advanced difficulty, Recommended category, 6–12h estimated effort. Note: related tests were computed where they served EDA business interpretation directly — a Kruskal-Wallis test across host segments and Mann-Whitney U tests comparing Superhost vs. non-Superhost listings (Section 4.4) — but these do not constitute the full, independently-documented hypothesis-testing framework Section 05 specifies.
6 - Data Science Challenges (ML)	Designed, not implemented	No models were trained. Price-prediction problem framing, feature-engineering design, and a 3-model comparison plan (OLS, Random Forest, XGBoost/LightGBM) were documented, but cross-validation, residual analysis, SHAP/LIME explainability, clustering/segmentation, and generalisation/bias testing were not executed.	Advanced difficulty, Optional category, 10–20h estimated effort — the largest single optional time block in the brief. Deprioritised below all mandatory and recommended sections.
7 - AI & Machine Learning Opportunities	Designed, not implemented	The assignment's own NLP/LLM deliverables — sentiment analysis, topic modeling, named entity recognition and a review-quality classifier on the 990,170-review corpus; a Retrieval-Augmented	Expert difficulty, Optional category, 8–16h estimated effort. Note: an LLM (Claude Sonnet 4.6) was used extensively as a coding and analysis assistant throughout the assignment — see the companion AI

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		Generation system and natural-language Q&A interface; a content-based recommendation engine; an AI-powered dynamic pricing advisor — were not built.	Usage Disclosure — but that is a different thing from building the NLP/LLM analytical features this section asks for.
8 - Open Innovation Challenge	Not attempted	No custom, self-directed product (interactive dashboard, AI analyst assistant, automated reporting system, novel analytical methodology, etc.) was built beyond the core deliverables.	Candidate-defined and fully optional. Given the one-week constraint, every hour available was directed at deepening the mandatory and recommended sections rather than building an additional, open-ended artifact.
12 -Cross-City Comparisons	Not applicable by design	No second city was ingested, so no comparative analytics, harmonisation work, or cross-market statistical correction (Bonferroni/FDR) could be produced.	This is the direct consequence of the single-city, depth-first strategy chosen in Section 2.2 (see Assumptions & Decisions Log, Section A.1) — not an oversight. The ingestion pipeline's CITY_CONFIG design means a second city could be added with a single dictionary entry, but this was not exercised within the one-week window.

What Would Be Done With More Time

Condensed from Section 14 (Future Improvements) of the main report, in rough priority order if the engagement continued:

Improvement	Why It Matters Most
Fix scraping failures (12 fully-null columns)	Unlocks host-level segmentation analysis (tenure, response rate, acceptance rate) that is currently completely blocked — the single highest-leverage data fix available.
Complete the formal H1–H5 statistical hypothesis-testing suite	Converts the indicative Section 05 results into fully verified statistics with confidence intervals and effect sizes, closing the largest gap between “designed” and “delivered.”
Train and validate the price-prediction model (OLS → Random Forest → XGBoost)	Delivers the Section 06 deliverable end-to-end, including SHAP explainability and bias/generalisation testing across neighbourhoods.
Full NLP pipeline on the 990K-review corpus	Sentiment analysis, topic modeling (BERTopic) and review-quality classification would unlock rich, currently-untapped demand-side intelligence.
Multi-scrape longitudinal tracking	A single monthly re-scrape removes the point-in-time snapshot limitation and enables true occupancy estimation and trend detection.
Multi-city pipeline deployment (10+ cities, containerised)	Exercises the CITY_CONFIG design that was built but not yet tested on a second city, and directly enables Section 12 cross-city comparisons.